

SYMPTOM REFERENCE GUIDE

Causes · Meanings · Consequences · Actions

30+ Common Symptoms | Disease Associations | When to Seek Help

Optimised for RAG / Vector-Store Ingestion

This document is for educational and RAG reference use only. It does not constitute medical advice. Always consult a qualified healthcare professional for diagnosis and treatment.

TABLE OF CONTENTS

PART A — PAIN SYMPTOMS

- A1. Body / Muscle Pain (Myalgia)
- A2. Joint Pain (Arthralgia)
- A3. Chest Pain
- A4. Back Pain
- A5. Headache
- A6. Migraine
- A7. Abdominal / Stomach Pain
- A8. Pelvic Pain
- A9. Nerve Pain (Neuropathy)

PART B — ENERGY & NEUROLOGICAL SYMPTOMS

- B1. Fatigue & Chronic Tiredness
- B2. Weakness (General & Muscular)
- B3. Low Energy / Lethargy
- B4. Brain Fog / Cognitive Impairment
- B5. Dizziness & Vertigo
- B6. Fainting (Syncope)
- B7. Numbness & Tingling (Paraesthesia)
- B8. Tremors & Involuntary Movements

PART C — DIGESTIVE SYMPTOMS

- C1. Stomach Upset / Nausea
- C2. Vomiting
- C3. Diarrhoea
- C4. Constipation
- C5. Bloating & Gas
- C6. Acid Reflux / Heartburn
- C7. Loss of Appetite
- C8. Jaundice

PART D — RESPIRATORY & CARDIOVASCULAR

- D1. Shortness of Breath (Dyspnoea)
- D2. Persistent Cough
- D3. Palpitations
- D4. High Blood Pressure (Hypertension)
- D5. Oedema (Swelling)

PART E — SYSTEMIC & SKIN SYMPTOMS

- E1. Fever

- E2. Night Sweats
- E3. Unexplained Weight Loss
- E4. Unexplained Weight Gain
- E5. Rash & Skin Changes
- E6. Hair Loss (Alopecia)
- E7. Excessive Thirst & Urination
- E8. Pale / Yellow Skin & Anaemia Signs

PART F — MENTAL & HORMONAL SYMPTOMS

- F1. Anxiety & Panic Attacks
- F2. Depression & Low Mood
- F3. Insomnia & Sleep Disturbance
- F4. Hormonal Imbalance Signs

PART G — SYMPTOM QUICK-REFERENCE TABLES

- G1. Symptom-to-Condition Mapping
- G2. Urgent vs. Non-Urgent Symptom Guide
- G3. Symptom Cluster Patterns

PART A — PAIN SYMPTOMS

A1. Body / Muscle Pain (Myalgia)

Myalgia refers to pain, aching, or soreness in one or more muscles. It is one of the most common complaints presenting to healthcare providers and can affect any muscle group in the body. It may be acute (days) or chronic (>3 months).

CAUSE

Overuse or exercise-induced micro-tears in muscle fibres (DOMS). Viral or bacterial infections (influenza, COVID-19, dengue fever — especially in tropical regions like Tamil Nadu). Autoimmune conditions (fibromyalgia, lupus, polymyositis). Electrolyte imbalances (magnesium, potassium, calcium deficiency). Hypothyroidism, adrenal insufficiency. Medication side effects (statins, ACE inhibitors, steroids). Dehydration. Stress and psychological tension causing sustained muscle contraction.

WHAT IT MEANS

Acute muscle pain after exertion is normal physiological repair. Generalised body pain with fever strongly suggests systemic infection. Persistent muscle pain without clear trigger warrants investigation for underlying systemic disease. Pain worsening at rest rather than with activity may indicate inflammatory or rheumatological cause.

CAN LEAD TO

Untreated chronic myalgia can lead to reduced mobility, muscle atrophy, depression, and sleep disruption. Underlying infectious causes (dengue, typhoid) can progress to organ complications if ignored. Rhabdomyolysis (extreme muscle breakdown) — rare but life-threatening — can cause acute kidney injury.

WHAT TO DO

Rest affected muscles; apply heat for chronic tension, ice for acute injury. Adequate hydration (2–3 litres/day). Electrolyte replacement. OTC analgesia (paracetamol/ibuprofen) for short-term relief. If fever accompanies — complete blood count (CBC), dengue NS1 antigen, malaria test. Persistent pain >2 weeks: thyroid profile, CK levels, ANA, ESR/CRP. Physiotherapy for chronic myofascial pain.

Characteristic	Likely Cause
Pain after gym/exercise, 24–48 hrs later	DOMS — normal, resolves in 3–5 days
Whole-body pain + fever + chills	Viral infection (flu, dengue, COVID-19)
Chronic widespread pain + fatigue + sleep issues	Fibromyalgia
Proximal muscle weakness + pain	Polymyositis / Dermatomyositis
Pain after starting statin medication	Statin-induced myopathy
Pain + cold intolerance + weight gain	Hypothyroidism
Pain worse in cold/damp + morning stiffness	Rheumatoid arthritis / Inflammatory

■ RED FLAGS — Seek Emergency Care Immediately:

- Severe muscle pain with dark/cola-coloured urine (rhabdomyolysis — kidney emergency).
- Sudden severe muscle weakness with difficulty swallowing or breathing.
- High fever + body pain + rash (possible dengue, meningococcal — emergency).
- Muscle pain with severe chest pain or jaw pain (possible cardiac event).

A2. Joint Pain (Arthralgia)

Arthralgia is pain in one or more joints without visible inflammation. Arthritis involves actual joint inflammation. Joint pain is extremely prevalent — affecting over 350 million people worldwide and is a leading cause of disability.

CAUSE

Osteoarthritis (wear and tear — most common, especially knees/hips). Rheumatoid arthritis (autoimmune — symmetric joint involvement). Gout (uric acid crystal deposition — classically big toe). Reactive arthritis following infection. Psoriatic arthritis. Lupus (SLE). Lyme disease (tick-borne). Viral arthritis (chikungunya — very common in South India/Chennai). Injury, bursitis, tendinitis.

WHAT IT MEANS

Acute joint pain after injury is usually structural damage. Symmetric morning stiffness >1 hour indicates autoimmune arthritis. Acute red hot swollen single joint (monoarthritis) indicates gout, septic arthritis, or reactive arthritis. Migratory joint pain (moving from joint to joint) suggests viral arthritis or rheumatic fever.

CAN LEAD TO

Untreated inflammatory arthritis leads to joint erosion, deformity, disability. Septic arthritis (infected joint) can destroy joint within days. Rheumatoid arthritis increases cardiovascular disease risk 2-fold. Untreated gout: kidney stones, tophi formation, chronic kidney disease.

WHAT TO DO

RICE for acute joint injuries. NSAIDs (ibuprofen) for inflammatory pain. Uric acid blood test if gout suspected; reduce purines (red meat, alcohol, organ meats). RF, anti-CCP, ANA, ESR/CRP if autoimmune suspected. X-ray/MRI for structural assessment. Rheumatology referral for persistent polyarthritis. Low-impact exercise (swimming, cycling) preserves joint function.

■ RED FLAGS — Seek Emergency Care Immediately:

- Single joint: red, hot, swollen, immovable with fever — possible septic arthritis (joint emergency).
- Joint pain in child with fever and rash — possible juvenile arthritis or rheumatic fever.
- Sudden severe big toe joint pain in night — acute gout attack, seek treatment.

A3. Chest Pain

Chest pain is one of the most important symptoms in medicine, ranging from benign causes (muscle strain, acid reflux) to immediately life-threatening conditions (heart attack, aortic dissection). All new unexplained chest pain should be taken seriously.

CAUSE

Cardiac: Angina (reduced blood flow to heart muscle), Myocardial infarction (heart attack), Pericarditis (sac inflammation). Respiratory: Pneumonia, Pleuritis, Pulmonary embolism, Pneumothorax. GI: GERD/acid reflux, Oesophageal spasm, Peptic ulcer. Musculoskeletal: Costochondritis, Intercostal muscle strain. Psychological: Panic attacks (very common and can mimic heart attack exactly). Shingles (herpes zoster — burning pain before rash).

WHAT IT MEANS

Cardiac chest pain: typically crushing, pressure-like, radiating to left arm/jaw/back, associated with sweating, breathlessness, nausea. Pleuritic pain: sharp, worse on breathing/coughing. GERD pain: burning, worse after meals and lying down, relieved by antacids. Panic attack: sharp/stabbing, with palpitations, tingling hands, fear of dying.

CAN LEAD TO

Untreated heart attack causes permanent heart muscle damage; mortality increases with every minute of delay. Untreated pulmonary embolism has 30% mortality. Aortic dissection is immediately fatal without surgery. Chronic GERD leads to Barrett's oesophagus and oesophageal cancer risk.

WHAT TO DO

Any new unexplained chest pain: ECG within 10 minutes. If cardiac suspected: chew 300 mg aspirin (if not allergic), call emergency services. Cardiac troponin blood test to rule out heart attack. Echo, stress test for stable angina evaluation. GERD: lifestyle modification, proton pump inhibitors (omeprazole). Physiotherapy for musculoskeletal chest pain.

■ RED FLAGS — Seek Emergency Care Immediately:

- **Crushing/pressure chest pain radiating to arm/jaw — call 112/999 immediately (possible heart attack).**
- **Sudden tearing chest/back pain (possible aortic dissection — surgical emergency).**
- **Chest pain with sudden breathlessness and one-sided lung silence (pneumothorax).**
- **Chest pain with leg swelling and recent long travel (possible pulmonary embolism).**

A4. Back Pain

Back pain is the single leading cause of disability worldwide. 80% of people experience significant back pain at some point in their lives. Most episodes are mechanical and resolve within 6 weeks; a small percentage have serious underlying causes requiring urgent investigation.

CAUSE

Mechanical (most common): Muscle/ligament strain, poor posture, disc herniation, facet joint arthritis. Structural: Spinal stenosis, spondylolisthesis, scoliosis. Inflammatory: Ankylosing spondylitis (young males, morning stiffness >3 months). Referred pain: Kidney stones, pyelonephritis, pancreatitis, aortic aneurysm. Serious: Spinal metastases (cancer spread), infection (discitis, epidural abscess), compression fracture (osteoporosis).

WHAT IT MEANS

Lower back pain worse with movement and better with rest = mechanical. Lower back pain worse at rest/night, better with movement = inflammatory (ankylosing spondylitis). Back pain with fever, night sweats, weight loss = infection or malignancy. Unilateral loin-to-groin pain with haematuria = kidney stone. Back pain with leg weakness/numbness = nerve compression.

CAN LEAD TO

Chronic untreated back pain leads to disability, opioid dependence, depression. Untreated cauda equina syndrome (nerve compression) = permanent paralysis and incontinence. Untreated spinal infection = sepsis, vertebral collapse. Missed spinal malignancy = delayed cancer treatment.

WHAT TO DO

Acute mechanical back pain: stay active (bed rest worsens outcome), NSAIDs, heat therapy. Physiotherapy and core strengthening exercises. MRI if neurological signs (leg weakness/numbness) or red flags. X-ray if structural deformity suspected. Refer for spinal MRI/specialist if not improving in 6 weeks. Ankylosing spondylitis: HLA-B27 test, rheumatology referral.

■ RED FLAGS — Seek Emergency Care Immediately:

- **Back pain with bladder/bowel incontinence or retention (cauda equina — spinal emergency).**
- **Back pain with fever >38.5°C and night sweats (possible spinal infection/abscess).**

- Back pain with progressive leg weakness bilaterally.
- Back pain in patient with known cancer history (possible metastasis).
- Back pain after significant trauma in elderly (possible fracture).

A5. Headache

Headache is among the most common neurological symptoms worldwide, affecting nearly everyone at some point. The International Headache Society classifies over 150 types. Most are primary (no underlying disease); a small percentage are secondary to serious conditions.

CAUSE

Primary: Tension-type (most common — band-like pressure, stress/posture related), Migraine (see A6), Cluster headache (severe unilateral, periorbital). Secondary: Hypertension (pressure/throbbing), Sinusitis (facial pressure), Meningitis (thunderclap + fever + neck stiffness), Subarachnoid haemorrhage (sudden worst headache of life), Brain tumour (progressive, worse in morning/lying down), Medication overuse headache (rebound from excess analgesic use).

WHAT IT MEANS

Bilateral pressure/tightness without nausea = tension headache. Throbbing unilateral with nausea, photophobia = migraine. Sudden-onset worst-ever headache = subarachnoid haemorrhage until proven otherwise. Progressive headache + morning vomiting + personality change = raised intracranial pressure. Headache + fever + neck stiffness + rash = meningitis.

CAN LEAD TO

Unmanaged tension/migraine headaches cause significant quality of life impairment and work loss. Medication overuse headache if analgesics taken >10–15 days/month. Missed hypertensive headache can precede hypertensive crisis or stroke. Untreated meningitis or SAH has high mortality.

WHAT TO DO

Tension headache: regular sleep, stress reduction, hydration, physiotherapy for cervicogenic. NSAIDs or paracetamol for acute episodes (not daily). Headache diary for migraine — identify and avoid triggers. BP measurement for all new onset headaches. CT head (non-contrast) if red flags present. Lumbar puncture if thunderclap headache with normal CT.

■ RED FLAGS — Seek Emergency Care Immediately:

- Sudden onset 'worst headache of life' — call emergency services (possible brain bleed).
- Headache + fever + stiff neck + rash — meningitis, call 112.
- Headache + visual changes + weakness/speech difficulty (possible stroke).
- Headache progressively worsening over weeks, worse lying down.
- New headache in patient with cancer or HIV.

A6. Migraine

Migraine is a complex neurological disorder characterised by recurrent moderate-to-severe headache, typically unilateral, pulsating, lasting 4–72 hours, and associated with nausea and sensitivity to light and sound. It affects ~15% of the global population and is the 2nd leading cause of disability worldwide. Women are 3x more affected than men. Approximately 30% of migraineurs experience aura.

CAUSE

Pathophysiology involves cortical spreading depression, trigeminal nerve sensitisation, and vasodilation of meningeal blood vessels. Triggers (individual): Hormonal fluctuations (menstruation), sleep changes, skipped meals, dehydration, alcohol (especially red wine), strong smells, bright lights, stress, tyramine-containing foods (aged cheese, processed meat), caffeine withdrawal, weather changes. Genetic predisposition is strong — 50% risk if both parents have migraine.

WHAT IT MEANS

A migraine attack represents abnormal brain hyperexcitability and neurovascular activation. Prodrome (hours before attack): mood changes, food cravings, yawning, fatigue. Aura (20–30 min before pain): visual zigzag lines, blind spots, tingling, speech difficulty. Headache phase: unilateral pulsating, worsened by activity, nausea, photo/phonophobia. Postdrome: exhaustion and cognitive impairment ('migraine hangover') lasting up to 24 hours.

CAN LEAD TO

Chronic migraine (>15 headache days/month) if not managed. Medication overuse headache (rebound headache) with frequent analgesic use. Migraine with aura increases stroke risk 2-fold, especially in women on oral contraceptives who smoke. Severe depression and anxiety are highly comorbid with migraine. Social isolation, job loss, relationship strain from frequent attacks.

WHAT TO DO

ACUTE TREATMENT: Triptans (sumatriptan) are first-line specific therapy — take early in attack. NSAIDs + antiemetics (domperidone/metoclopramide) for moderate attacks. Rest in dark, quiet room; cold compress to head. Hydration. PREVENTION (if >4 attacks/month): Beta-blockers (propranolol), Topiramate, Amitriptyline, CGRP antagonists (newer targeted therapy). Magnesium 400 mg/day, Riboflavin B2 400 mg/day have evidence as preventive supplements. Maintain headache diary to identify personal triggers. Regular sleep, meals, and exercise schedule.

Migraine Phase	Duration	Symptoms	Management
Prodrome	Hours to days	Mood changes, cravings, fatigue, yawning	Identify — prepare medication
Aura	5–60 minutes	Visual zigzags, numbness, speech difficulty	Take triptan/NSAID early
Headache	4–72 hours	Pulsating unilateral pain, nausea, vomiting, photophobia	Triptan + dark rest
Postdrome	Up to 24 hours	Brain fog, exhaustion, irritability	Rest, gentle hydration

■ RED FLAGS — Seek Emergency Care Immediately:

- First ever worst headache of life — subarachnoid haemorrhage must be excluded.
- Aura lasting >60 minutes or with new neurological deficits — possible stroke.
- Migraine aura without headache in a patient on OCP — increased stroke risk, discuss with doctor.
- Fever + neck stiffness accompanying headache — meningitis.
- Headache not responding to any treatment and progressively worsening.

A7. Abdominal / Stomach Pain

Abdominal pain is one of the most frequent presenting complaints in medicine. Location, character, timing, and associated symptoms are crucial for identifying the cause. The abdomen contains organs of the digestive, urinary, reproductive, and vascular systems.

CAUSE

Upper central (epigastric): Gastritis, peptic ulcer, GERD, pancreatitis, MI. Right upper quadrant: Gallstones, cholecystitis, hepatitis, liver abscess. Left upper quadrant: Splenomegaly, splenic infarct. Right lower quadrant: Appendicitis (most important), ovarian cyst, inguinal hernia. Left lower quadrant: Diverticulitis, IBS, ovarian cyst. Central/umbilical: Early appendicitis, small bowel obstruction, mesenteric ischaemia. Generalised: Peritonitis, gastroenteritis, IBS, constipation.

WHAT IT MEANS

Colicky pain (comes and goes in waves) = hollow organ obstruction (gallstone, kidney stone, bowel obstruction). Constant dull aching = solid organ inflammation (pancreatitis, hepatitis, appendicitis). Sharp burning epigastric pain worsened by fasting and improved by food = peptic ulcer. Right lower quadrant pain migrating from umbilicus with fever/nausea = appendicitis. Lower abdominal cramping relieved by defecation = IBS.

CAN LEAD TO

Untreated appendicitis: perforation, peritonitis, sepsis — mortality rises dramatically. Untreated peptic ulcer: perforation, GI bleeding. Chronic gastritis with H. pylori: gastric cancer risk. Untreated bowel obstruction: bowel ischaemia and necrosis. Pancreatitis: organ failure, chronic pancreatitis, diabetes.

WHAT TO DO

Assessment: SOCRATES (Site, Onset, Character, Radiation, Alleviating/Aggravating, Timing, Severity). Blood tests: FBC, LFT, amylase/lipase, CRP. Urine dipstick. Ultrasound abdomen: gallstones, liver, kidneys, appendix. CT abdomen/pelvis if surgical cause suspected. NPO (nothing by mouth) and IV fluids if surgical emergency. H. pylori test (stool antigen / urea breath test) and eradication therapy if positive. Dietary modifications for IBS: low FODMAP diet.

■ RED FLAGS — Seek Emergency Care Immediately:

- Sudden severe abdominal pain with board-like rigidity — peritonitis, surgical emergency.
- Abdominal pain + vomiting + complete obstipation (no stool/gas) — bowel obstruction.
- RLQ pain + fever + anorexia — possible appendicitis — seek immediate surgical review.
- Severe epigastric pain radiating to back + elevated amylase — acute pancreatitis.
- Abdominal pain with haematemesis (vomiting blood) or melaena (black tarry stool).
- Ruptured ectopic pregnancy: severe lower abdominal pain in women of childbearing age.

A8. Pelvic Pain

Pelvic pain refers to pain in the lower abdomen and pelvis. It is more prevalent in women due to gynaecological causes but also affects men. It can be acute or chronic (>6 months).

CAUSE

Women: Menstrual cramps (dysmenorrhoea), ovarian cysts, endometriosis, pelvic inflammatory disease (PID), ectopic pregnancy, uterine fibroids, PCOS. Men: Prostatitis, epididymo-orchitis. Both: Urinary tract infection (UTI), kidney stones, IBS, appendicitis, hernia, interstitial cystitis, musculoskeletal dysfunction.

WHAT IT MEANS

Cyclical pain with menstruation = dysmenorrhoea or endometriosis. Pain with abnormal vaginal discharge and fever = PID (sexually transmitted infection complication). Sudden unilateral pain in early pregnancy = ectopic pregnancy until proven otherwise. Burning pain with urinary frequency = UTI.

CAN LEAD TO

Untreated endometriosis: infertility, adhesions, chronic pain. Untreated PID: tubal scarring, ectopic pregnancy risk, infertility. Ruptured ectopic pregnancy: massive haemorrhage, maternal death. Untreated ovarian cyst: torsion — loss of ovary.

WHAT TO DO

Pregnancy test in all women of childbearing age with pelvic pain. Pelvic ultrasound. Vaginal swabs for STI/PID. Urine culture. Gynaecology referral for chronic pelvic pain. Laparoscopy may be needed for definitive endometriosis diagnosis. Pain management: NSAIDs, hormonal therapy for endometriosis.

■ RED FLAGS — Seek Emergency Care Immediately:

- Sudden severe one-sided pelvic pain in pregnancy — possible ectopic rupture, call 112.
- Pelvic pain + fever + foul discharge — PID, start antibiotics urgently.
- Ovarian torsion: sudden severe unilateral pain with nausea/vomiting — surgical emergency.

A9. Nerve Pain (Neuropathic Pain / Neuropathy)

Neuropathic pain arises from damage or dysfunction of the nervous system itself. Unlike nociceptive pain (from tissue damage), nerve pain is often described as burning, shooting, electric-shock, or stabbing sensations and may occur spontaneously without a clear stimulus.

CAUSE

Peripheral neuropathy: Diabetes (most common cause globally), alcohol, B12 deficiency, HIV, chemotherapy, hypothyroidism, kidney disease, autoimmune (Guillain-Barre). Entrapment neuropathies: Carpal tunnel (median nerve), sciatica (lumbar disc). Post-herpetic neuralgia (after shingles). Central neuropathic pain: Multiple sclerosis, stroke, spinal cord injury.

WHAT IT MEANS

Burning feet and hands = classic diabetic peripheral neuropathy. Shooting leg pain from back = sciatica from disc herniation. Wrist pain/tingling at night = carpal tunnel syndrome. Burning pain following a dermatomal (band-like) distribution after rash = post-herpetic neuralgia.

CAN LEAD TO

Untreated diabetic neuropathy: foot ulcers, infections, amputations. Progressive neuropathy leads to loss of protective sensation, falls, disability. Carpal tunnel: permanent nerve damage if untreated.

WHAT TO DO

Fasting glucose, HbA1c for diabetes. B12 levels. TFTs. Renal function. Nerve conduction studies (NCS/EMG) for assessment. Optimise blood glucose control in diabetics. Medications: Pregabalin, Gabapentin, Duloxetine, Amitriptyline for neuropathic pain. B12 injections/supplements if deficient. Carpal tunnel: wrist splint, steroid injection, surgical release.

■ RED FLAGS — Seek Emergency Care Immediately:

- Rapidly progressive bilateral weakness and numbness (possible Guillain-Barre — emergency).
- Nerve pain with urinary/bowel changes (possible spinal cord compression).

- Painful neuropathy with unintentional weight loss (possible underlying malignancy).

PART B — ENERGY & NEUROLOGICAL SYMPTOMS

B1. Fatigue & Chronic Tiredness

Fatigue is an overwhelming sense of tiredness, lack of energy, and exhaustion that is not relieved by rest. It is one of the most common presenting complaints and can be a symptom of almost any illness. Chronic fatigue is defined as persisting >6 months.

CAUSE

Lifestyle: Sleep deprivation, sedentary behaviour, poor diet, overtraining. Medical: Anaemia (iron deficiency most common), hypothyroidism, diabetes, chronic kidney/liver disease, heart failure, COPD. Infections: Post-viral fatigue (COVID-19 long-haul, Epstein-Barr / glandular fever). Mental health: Depression (fatigue in >90% of depressed patients), anxiety, burnout. Autoimmune: Lupus, rheumatoid arthritis. Nutritional: B12, folate, vitamin D, iron deficiency. Sleep disorders: Obstructive sleep apnoea, insomnia. Medications: Beta-blockers, antihistamines, benzodiazepines, statins.

WHAT IT MEANS

Fatigue that improves with rest and has an identifiable cause is usually secondary fatigue. Fatigue unresponsive to rest, present for months, with post-exertional malaise = Myalgic Encephalomyelitis/Chronic Fatigue Syndrome (ME/CFS). Fatigue with low mood, anhedonia, and sleep changes = depression. Fatigue with pallor, shortness of breath, palpitations = anaemia. Fatigue with weight gain, cold intolerance, constipation = hypothyroidism.

CAN LEAD TO

Unaddressed fatigue causes reduced work performance, social withdrawal, accidents, and injury risk. Underlying anaemia: cardiac strain, impaired fetal development in pregnancy. Untreated sleep apnoea: hypertension, arrhythmias, cognitive decline, increased accident risk. Depression if untreated: worsening mental health, suicide risk.

WHAT TO DO

Comprehensive blood work: CBC (anaemia), TSH (thyroid), fasting glucose, ferritin, B12/folate, vitamin D, CRP, LFT, renal function, cortisol if adrenal issues suspected. Sleep hygiene assessment; polysomnography if sleep apnoea suspected (snoring + daytime sleepiness). Mental health screening (PHQ-9 for depression). Graded exercise therapy (carefully titrated in ME/CFS). Nutritional deficiency correction. Review and modify medications.

■ RED FLAGS — Seek Emergency Care Immediately:

- Fatigue with unexplained weight loss, night sweats, lymphadenopathy — possible malignancy.
- Fatigue with shortness of breath at rest and leg swelling — possible heart failure.
- Severe fatigue with adrenal crisis signs (hypotension, vomiting, confusion) — Addison's emergency.

B2. Weakness (General & Muscular)

Weakness refers to reduced ability to exert force with muscles. It must be distinguished from fatigue (tiredness without true loss of strength) and from perceived weakness. True weakness indicates neuromuscular pathology.

CAUSE

Neuromuscular: Stroke (sudden unilateral), Motor neurone disease (ALS), Myasthenia gravis (fatigable weakness), Guillain-Barre syndrome (ascending weakness). Metabolic: Hypokalaemia (low potassium), hypocalcaemia, hypoglycaemia, hypothyroidism. Muscle disease: Polymyositis, muscular dystrophies, rhabdomyolysis. Spinal cord: Compression, transverse myelitis. Medications: Steroids (proximal myopathy), statins.

WHAT IT MEANS

Sudden unilateral weakness (face/arm/leg) = stroke until proven otherwise. Proximal weakness (difficulty standing from chair, raising arms) = myopathy or myositis. Distal weakness (foot drop, weak grip) = peripheral neuropathy. Fatigable weakness (worse with use, improves with rest) = Myasthenia gravis (ptosis + diplopia key clue). Ascending paralysis after infection = Guillain-Barre syndrome.

CAN LEAD TO

Untreated stroke: permanent disability, death. GBS without respiratory monitoring: respiratory failure. Progressive untreated MND: paralysis, respiratory failure. Untreated hypokalaemia: cardiac arrhythmias.

WHAT TO DO

STROKE: FAST test (Face drooping, Arm weakness, Speech difficulty, Time to call 112). Electrolytes for metabolic causes. CK enzyme for muscle disease. Nerve conduction studies. MRI brain/spine. Edrophonium/Tensilon test or antibody tests for Myasthenia gravis.

■ RED FLAGS — Seek Emergency Care Immediately:

- Sudden-onset unilateral weakness + facial droop + speech problems — STROKE, call 112 NOW.
- Rapidly ascending bilateral leg weakness (GBS) — ICU monitoring needed.
- Weakness with difficulty breathing or swallowing — respiratory emergency.

B3. Low Energy / Lethargy

Lethargy is a state of tiredness, sluggishness, and mental dullness that is less severe than fatigue but significantly impacts daily functioning. It represents reduced drive, motivation, and vitality.

CAUSE

Poor sleep quality or quantity. Nutritional deficiencies (iron, B12, vitamin D, magnesium). Hormonal: Low testosterone (hypogonadism), oestrogen deficiency, thyroid dysfunction. Adrenal insufficiency. High cortisol (chronic stress / Cushing's). Sedentary lifestyle and deconditioning. Chronic pain. Dehydration. Excess carbohydrate diet causing blood sugar spikes and crashes. Medications: Beta-blockers, antidepressants, antiepileptics.

WHAT IT MEANS

Low energy in the morning improving through day = poor sleep quality. Energy crash after meals = blood sugar dysregulation or reactive hypoglycaemia. Persistent low energy + depression + weight changes = thyroid or hormonal issue. Low energy + excessive thirst/urination = diabetes.

CAN LEAD TO

Chronic low energy leads to metabolic syndrome, weight gain, cardiovascular risk. Hormonal deficiencies untreated affect bone density, cardiovascular health, cognition.

WHAT TO DO

Blood tests: thyroid panel, testosterone, cortisol, full metabolic panel, vitamin levels. Sleep assessment. Dietary review — blood sugar optimisation (reduce refined carbs, increase protein). Regular aerobic exercise paradoxically improves energy. Hormone replacement therapy where indicated (under specialist guidance).

■ **RED FLAGS — Seek Emergency Care Immediately:**

- **Sudden severe lethargy, confusion, extreme thirst in known diabetic — possible ketoacidosis.**
- **Lethargy with severe hypotension, vomiting, darkening skin — Addison's disease crisis.**

B4. Brain Fog / Cognitive Impairment

Brain fog is not a medical diagnosis but describes a cluster of cognitive symptoms: difficulty concentrating, poor memory, slow thinking, mental fatigue, and confusion. It significantly impacts quality of life and occupational functioning.

CAUSE

Post-viral: Long COVID, post-Epstein-Barr (glandular fever). Nutritional: B12 deficiency (critical for myelin), folate, iron, omega-3 deficiency. Hormonal: Hypothyroidism, menopause (oestrogen decline), testosterone deficiency. Sleep deprivation or sleep apnoea. Mental health: Depression, anxiety, ADHD. Medications: Antihistamines, benzodiazepines, anticholinergics. Inflammatory: Autoimmune conditions (lupus, multiple sclerosis). Blood sugar: Hypoglycaemia, poorly controlled diabetes. Gut-brain axis dysbiosis.

WHAT IT MEANS

Brain fog that started post-COVID and includes fatigue and breathlessness = Long COVID. Brain fog + weight gain + cold intolerance = hypothyroidism. Cognitive decline in elderly with memory loss, personality changes = early dementia. Cognitive symptoms with mood disorder = depression-related cognitive impairment.

CAN LEAD TO

Untreated B12 deficiency: irreversible spinal cord and brain damage. Unmanaged depression: cognitive worsening, social and occupational decline. Untreated sleep apnoea: accelerated cognitive decline and dementia risk.

WHAT TO DO

B12, folate, vitamin D, thyroid function, fasting glucose, CBC. Cognitive assessment tools (MoCA, MMSE) in elderly. Sleep study. Mental health assessment. Omega-3 supplementation. Gut microbiome optimisation. Structured sleep, exercise, and stress management.

■ **RED FLAGS — Seek Emergency Care Immediately:**

- **Rapid-onset cognitive impairment with fever/headache — possible encephalitis.**
- **Progressive memory loss with personality changes in young person — neurological review.**
- **Confusion + focal neurological signs + history of cancer — brain metastasis possible.**

B5. Dizziness & Vertigo

Dizziness is a broad term encompassing vertigo (spinning sensation), presyncope (lightheadedness), and disequilibrium (balance problems). Accurately characterising which type is essential for diagnosis.

CAUSE

Vertigo: BPPV (Benign Paroxysmal Positional Vertigo — most common), Labyrinthitis/vestibular neuritis (post-viral), Meniere's disease (tinnitus + ear fullness), Central vertigo (brainstem/cerebellar stroke or tumour). Presyncope: Orthostatic hypotension (standing up too fast), dehydration, cardiac arrhythmias, anaemia, hypoglycaemia, vasovagal. Chronic dizziness: anxiety, medication side effects, cervicogenic.

WHAT IT MEANS

Brief positional vertigo triggered by head movement = BPPV (very common, very treatable). Severe prolonged vertigo with nausea/vomiting after viral illness = vestibular neuritis. Episodic vertigo with fluctuating hearing loss and tinnitus = Meniere's disease. Sudden vertigo + vomiting + difficulty walking + headache = cerebellar stroke.

CAN LEAD TO

BPPV and vestibular disorders: falls, injury, especially in elderly. Untreated Meniere's: progressive hearing loss. Missed posterior fossa stroke: permanent disability or death.

WHAT TO DO

Epley manoeuvre for BPPV (highly effective — over 90% cure rate). Vestibular rehabilitation for chronic vestibular disorders. Orthostatic hypotension: hydration, rise slowly, compression stockings. ECG, Holter monitor for arrhythmia-related dizziness. MRI brain if central cause suspected.

■ RED FLAGS — Seek Emergency Care Immediately:

- Sudden severe vertigo + inability to walk + vomiting + headache — possible cerebellar stroke.
- Dizziness with hearing loss in one ear suddenly — possible viral/ischaemic emergency.
- Dizziness with palpitations or syncope — cardiac arrhythmia evaluation needed urgently.

PART C — DIGESTIVE SYMPTOMS

C1. Stomach Upset / Nausea

Nausea is the unpleasant sensation of impending vomiting. It is a non-specific symptom arising from GI, neurological, metabolic, or psychological sources and is mediated via the vomiting centre in the brainstem.

CAUSE

Gastrointestinal: Gastroenteritis (most common), gastritis, GERD, peptic ulcer, food poisoning, IBS, gastroparesis (delayed stomach emptying — common in diabetes). Non-GI: Pregnancy (morning sickness), medications (antibiotics, NSAIDs, iron, chemotherapy), migraine, labyrinthitis, anxiety/panic, uraemia (kidney failure), liver failure, raised intracranial pressure, myocardial infarction.

WHAT IT MEANS

Nausea after contaminated food/water with diarrhoea = food poisoning or gastroenteritis. Morning nausea in women of reproductive age = pregnancy until proven otherwise. Nausea + heartburn + bloating = GERD or gastritis. Nausea + severe headache = migraine or raised ICP. Nausea + chest pain = MI warning.

CAN LEAD TO

Persistent nausea: dehydration, electrolyte imbalances, malnutrition. Untreated gastroparesis: poor diabetes control, malnutrition. Nausea as symptom of underlying serious disease if missed.

WHAT TO DO

Oral rehydration therapy for gastroenteritis. Small, frequent bland meals (BRAT: bananas, rice, applesauce, toast). Antiemetics: ondansetron, domperidone, metoclopramide. Pregnancy test if indicated. Medication review. Endoscopy if nausea associated with heartburn, weight loss, or >4 weeks.

■ RED FLAGS — Seek Emergency Care Immediately:

- Nausea + projectile vomiting + severe headache + neck stiffness — meningitis/ICP.
- Nausea + crushing chest pain/sweating — possible MI.
- Nausea with signs of severe dehydration (sunken eyes, no urine, altered consciousness).

C2. Vomiting

Vomiting (emesis) is the forceful expulsion of gastric contents. It must be distinguished from regurgitation (effortless return of food) and rumination. The content, colour, and timing of vomiting provide important diagnostic clues.

CAUSE

Gastroenteritis, food poisoning (Salmonella, Staph aureus, Bacillus cereus). Surgical: appendicitis, bowel obstruction, intussusception. Metabolic: DKA (diabetic ketoacidosis), uraemia, Addison's crisis. Central: raised ICP, meningitis, posterior fossa tumours. Cyclic vomiting syndrome. Bulimia nervosa. Pregnancy (hyperemesis gravidarum).

WHAT IT MEANS

Vomit with bile (yellow/green) = pyloric obstruction is above the blockage. Faeculent (faecal-smelling) vomit = distal bowel obstruction. Haematemesis (blood in vomit) = upper GI bleed. Projectile vomiting in infant = pyloric stenosis.

CAN LEAD TO

Dehydration, hypokalaemia, metabolic alkalosis. Aspiration pneumonia (especially in elderly or unconscious). Mallory-Weiss tear (oesophageal tear from repeated vomiting causing haematemesis).

WHAT TO DO

IV fluid replacement if severe. Antiemetics. Correct electrolytes. CT abdomen if obstruction suspected. Endoscopy for haematemesis. NPO if surgical cause suspected.

■ RED FLAGS — Seek Emergency Care Immediately:

- Vomiting blood (haematemesis) — upper GI bleed, emergency.
- Vomiting with inability to pass stool/gas and abdominal distension — bowel obstruction.
- Persistent vomiting in diabetic with fruity breath — possible DKA.

C3. Diarrhoea

Diarrhoea is defined as passage of 3 or more loose/watery stools in 24 hours. Acute diarrhoea (<2 weeks) is usually infectious; chronic diarrhoea (>4 weeks) requires investigation. Globally responsible for 1.6 million deaths/year, primarily in children.

CAUSE

Infectious: Viral (rotavirus, norovirus — most common), bacterial (Salmonella, Shigella, E. coli, Campylobacter, Vibrio cholerae), parasitic (Giardia, Entamoeba). Non-infectious: IBS, IBD (Crohn's, ulcerative colitis), coeliac disease, lactose intolerance, food allergy, medications (antibiotics — C. diff colitis), hyperthyroidism, bile acid malabsorption, colorectal cancer.

WHAT IT MEANS

Watery diarrhoea without blood = viral gastroenteritis or toxin-mediated. Bloody diarrhoea (dysentery) = invasive bacterial infection (Shigella, E. coli 0157) or IBD. Diarrhoea alternating with constipation = IBS or colorectal cancer. Nocturnal diarrhoea (waking from sleep) = almost always organic disease, not IBS.

CAN LEAD TO

Severe dehydration, especially in elderly and children. Haemolytic uraemic syndrome (E. coli 0157): kidney failure. Untreated IBD: surgery, malnutrition, colorectal cancer. C. diff colitis: toxic megacolon.

WHAT TO DO

ORS (oral rehydration salts) — mainstay of treatment. IV fluids if severe. Stool culture and microscopy if bloody or prolonged. AVOID loperamide (Imodium) in bloody diarrhoea or if infection suspected. Colonoscopy for chronic diarrhoea. Coeliac antibodies (anti-TTG IgA). Specific antibiotics only for identified bacterial causes (avoid empirical antibiotics).

■ RED FLAGS — Seek Emergency Care Immediately:

- Bloody diarrhoea with high fever — possible dysentery or IBD flare, seek care.
- Diarrhoea with severe dehydration signs (no urine, sunken eyes, confusion).
- Diarrhoea post-antibiotic course — possible C. difficile colitis.
- Rice-water stools in large volume rapidly — possible cholera in endemic areas.

C4. Constipation

Constipation is defined as fewer than 3 bowel movements per week, or difficult/painful defecation with hard stools. It affects up to 20% of the global population and is more prevalent in women and elderly. Functional constipation is

most common, but secondary causes must be excluded.

CAUSE

Lifestyle: Low fibre diet, inadequate hydration, sedentary behaviour, ignoring urge to defecate. Medical: Hypothyroidism, diabetes (autonomic neuropathy), hypercalcaemia, hypokalaemia, Parkinson's disease, multiple sclerosis. Structural: Colorectal cancer, stricture, diverticular disease, pelvic floor dysfunction. Medications: Opioids (most common drug-induced cause), iron supplements, calcium channel blockers, anticholinergics, aluminium antacids.

WHAT IT MEANS

Constipation in an otherwise healthy young person with dietary triggers = functional. New onset constipation in person >50 years = must rule out colorectal cancer. Constipation alternating with diarrhoea + bloating = IBS. Constipation + weight gain + fatigue + cold intolerance = hypothyroidism.

CAN LEAD TO

Faecal impaction — massive stool accumulation requiring manual evacuation. Haemorrhoids, anal fissures from straining. Overflow diarrhoea (liquid stool leaking past impaction — misdiagnosed as diarrhoea in elderly). Sigmoid volvulus in severe chronic constipation.

WHAT TO DO

Increase dietary fibre gradually to 25–38 g/day (fruits, vegetables, legumes, whole grains). Hydration: minimum 2 litres/day. Regular physical activity. Laxatives: Osmotic (macrogol/polyethylene glycol — first choice), Stimulant (senna — short-term), Bulk-forming (ispaghula/psyllium — gentle, long-term). Colonoscopy if new onset >50, blood in stool, family history of colorectal cancer. Thyroid function, calcium levels. Review and change causative medications where possible.

■ RED FLAGS — Seek Emergency Care Immediately:

- New onset constipation >50 years with blood in stool or weight loss — colonoscopy urgent.
- Constipation with abdominal distension and absolute constipation (no flatus) — obstruction.
- Constipation in child with ribbon-like stools and poor growth — possible Hirschsprung's disease.

C5. Bloating & Excessive Gas

Bloating is the sensation of abdominal fullness or distension. It is common, affecting up to 30% of the population, and is usually benign but can indicate underlying GI pathology.

CAUSE

Dietary: Gas-producing foods (beans, lentils, cruciferous vegetables, onions, carbonated drinks). Aerophagia (swallowing air — anxiety, eating fast, chewing gum). SIBO (Small Intestinal Bacterial Overgrowth). Lactose/fructose intolerance. IBS (most common cause of chronic bloating). Coeliac disease. IBD, gastroparesis, ascites (fluid in abdomen — liver disease), ovarian cancer.

WHAT IT MEANS

Bloating after specific foods (dairy, wheat, legumes) = food intolerance. Constant bloating worsening through the day with altered bowel habit = IBS or SIBO. Bloating with weight gain and distended abdomen in older person = possible ascites or malignancy. Bloating with early satiety and weight loss = gastric cancer red flag.

CAN LEAD TO

Coeliac if untreated: malabsorption, osteoporosis, increased bowel cancer risk. Untreated SIBO: nutritional deficiencies, leaky gut. Ascites from liver cirrhosis: infection (SBP), hepatic encephalopathy.

WHAT TO DO

Food and symptom diary to identify triggers. Low-FODMAP diet trial for IBS. Lactose breath test. Hydrogen breath test for SIBO. Anti-TTG IgA for coeliac. Ultrasound for ascites. Probiotics may reduce bloating in IBS. Rifaximin for SIBO.

■ RED FLAGS — Seek Emergency Care Immediately:

- Rapidly increasing abdominal girth with pain — possible ascites or obstruction.
- Bloating with early satiety, weight loss in older adult — upper GI malignancy.
- Bloating + severe pain + fever + rigidity — peritonitis.

C6. Acid Reflux / Heartburn (GERD)

Gastro-oesophageal reflux disease (GERD) occurs when stomach acid flows back into the oesophagus, causing burning discomfort. It affects approximately 20% of Western populations and is increasing globally.

CAUSE

Lower oesophageal sphincter (LES) dysfunction — allows acid to reflux. Hiatal hernia (part of stomach slides into chest). Obesity (increased abdominal pressure). Pregnancy. Dietary triggers: spicy food, fatty food, chocolate, coffee, alcohol, citrus. Smoking (reduces LES tone). Delayed gastric emptying. NSAIDs and aspirin (damage gastric mucosa). H. pylori infection.

WHAT IT MEANS

Burning sensation in chest/throat rising after meals, especially when lying down = classic GERD. Sour/acid taste in mouth = acid reaching oropharynx. Chronic cough, hoarseness, or asthma worsening = extra-oesophageal reflux.

CAN LEAD TO

Oesophagitis (inflammation). Barrett's oesophagus (pre-cancerous change) in 10–15%. Oesophageal adenocarcinoma — fastest rising cancer in many countries. Dental erosion. Laryngitis. Aspiration pneumonia.

WHAT TO DO

Lifestyle: Elevate head of bed, avoid eating 3 hrs before bed, weight loss, quit smoking. Antacids for immediate relief. H2 blockers (ranitidine/famotidine). Proton Pump Inhibitors (omeprazole, pantoprazole — most effective) — use minimum effective dose. Upper GI endoscopy if >40 years, alarm symptoms, or not responding to PPIs. H. pylori test and treat if positive.

■ RED FLAGS — Seek Emergency Care Immediately:

- Difficulty swallowing (dysphagia) especially solids — possible oesophageal cancer.
- GERD with unexplained weight loss — urgent endoscopy.
- Haematemesis or melaena with reflux symptoms — upper GI bleed.

PART D — RESPIRATORY & CARDIOVASCULAR SYMPTOMS

D1. Shortness of Breath (Dyspnoea)

Dyspnoea is the subjective sensation of breathlessness that is disproportionate to the level of physical exertion. It represents a mismatch between the demand for ventilation and the ability to deliver it and can arise from cardiac, pulmonary, haematological, or metabolic causes.

CAUSE

Cardiac: Heart failure, acute MI, arrhythmia, cardiac tamponade. Pulmonary: Asthma, COPD, pneumonia, pulmonary embolism (PE), pneumothorax, pulmonary fibrosis. Haematological: Severe anaemia. Metabolic: Diabetic ketoacidosis (Kussmaul breathing), severe acidosis. Psychiatric: Anxiety/panic disorder (hyperventilation syndrome). Chest wall: Massive pleural effusion, obesity hypoventilation.

WHAT IT MEANS

Sudden-onset breathlessness with pleuritic chest pain + risk factors (DVT, recent surgery) = PE. Breathlessness at rest, worse lying flat (orthopnoea), with ankle swelling = heart failure. Episodic wheeze and breathlessness triggered by allergens/exercise = asthma. Breathlessness with productive cough and fever = pneumonia. Breathlessness with tingling hands/lips + anxiety = hyperventilation.

CAN LEAD TO

Untreated heart failure: progressive deterioration, hospitalisation, death. Untreated PE: haemodynamic collapse and cardiac arrest (mortality 30% if untreated). Uncontrolled asthma: status asthmaticus — life-threatening. Untreated COPD: respiratory failure.

WHAT TO DO

Pulse oximetry (SpO₂) — target >94%. ECG, chest X-ray, BNP/NT-proBNP (heart failure marker), D-dimer (PE screening). ABG (arterial blood gas) in severe cases. CTPA for PE. Echocardiogram for cardiac causes. Salbutamol inhaler for asthma/COPD. Diuretics for heart failure. Anticoagulation for PE.

■ RED FLAGS — Seek Emergency Care Immediately:

- SpO₂ <90%, severe breathlessness at rest — immediate emergency care.
- Sudden onset breathlessness with chest pain — PE or pneumothorax.
- Breathlessness with stridor (high-pitched noise) — upper airway obstruction.
- Breathlessness in MI setting — acute pulmonary oedema.

D2. Persistent Cough

Cough is a vital protective reflex that clears airways. Acute cough <3 weeks; subacute 3–8 weeks; chronic cough >8 weeks. Chronic cough affects 10% of the population and has significant impact on quality of life.

CAUSE

Acute: URTI (viral — most common), pneumonia, acute bronchitis, foreign body. Chronic: Post-nasal drip (UACS) — most common; GERD — 2nd most common; Asthma (cough-variant); ACE inhibitor drugs (dry cough in 10–15%); Eosinophilic bronchitis; COPD; Bronchiectasis; Pulmonary fibrosis; Lung cancer; Tuberculosis (especially in India/South Asia — must consider).

WHAT IT MEANS

Dry irritating cough on ACE inhibitor = drug side effect (switch to ARB). Cough worse when lying down + heartburn = GERD-related cough. Nocturnal cough with wheeze = asthma. Productive cough with fever/purulent sputum = LRTI/pneumonia. Chronic productive cough in smoker + breathlessness = COPD. Haemoptysis (coughing blood) = serious — TB, cancer, PE.

CAN LEAD TO

Untreated TB: disease progression, transmission, MDR-TB. Undiagnosed lung cancer: poor prognosis if late-stage. Chronic GERD-induced cough: oesophageal complications. Uncontrolled asthma: reduced lung function.

WHAT TO DO

Chest X-ray for any cough >3 weeks. Sputum AFB smear x3 if TB suspected (cough >2 weeks + fever + night sweats in India). Spirometry for asthma/COPD. Skin-prick testing for allergic causes. CT chest if CXR normal and cough persistent. Review medications for ACE inhibitors.

■ RED FLAGS — Seek Emergency Care Immediately:

- Haemoptysis (coughing blood) with or without weight loss — urgent chest CT/bronchoscopy.
- Cough + night sweats + weight loss + fever in South Asia — TB screening is mandatory.
- Stridor (croup-like) or complete loss of voice with cough — airway compromise.

D3. Palpitations

Palpitations are the awareness of one's own heartbeat, perceived as fast, irregular, pounding, or fluttering. They are common and often benign but can signal life-threatening arrhythmias.

CAUSE

Benign: Sinus tachycardia (anxiety, caffeine, exercise, fever, anaemia, thyrotoxicosis). Ectopic beats (PACs/PVCs — very common, often harmless). Arrhythmias: AF (atrial fibrillation — most common significant arrhythmia), SVT (supraventricular tachycardia), VT (ventricular tachycardia — dangerous), WPW syndrome. Structural: Valvular disease, cardiomyopathy. Metabolic: Hyperthyroidism, hypoglycaemia, electrolyte imbalances.

WHAT IT MEANS

Occasional missed beats followed by a thump = ectopic beats (benign in structurally normal heart). Sudden onset regular rapid heartbeat (150–250 bpm) that stops suddenly = SVT. Completely irregular heartbeat = AF (stroke risk — needs anticoagulation assessment). Palpitations with syncope or presyncope = potentially malignant arrhythmia.

CAN LEAD TO

Untreated AF: 5-fold increased stroke risk (thrombus formation in left atrial appendage). Untreated VT: ventricular fibrillation and cardiac arrest. Untreated hyperthyroidism-driven tachycardia: thyroid crisis (thyroid storm).

WHAT TO DO

12-lead ECG — essential first test. 24-hour/48-hour Holter monitor to capture intermittent arrhythmias. Event recorder for infrequent palpitations. Echocardiogram for structural heart disease. TFTs, FBC, electrolytes, glucose. CHA2DS2-VASc score for AF anticoagulation decision.

■ RED FLAGS — Seek Emergency Care Immediately:

- Palpitations with syncope, presyncope, or near-collapse — cardiac emergency.
- Palpitations with chest pain or breathlessness — rule out MI and serious arrhythmia.

- Sustained rapid irregular heartbeat >30 minutes — emergency ECG.

PART E — SYSTEMIC & SKIN SYMPTOMS

E1. Fever

Fever is defined as a core body temperature $>38.0^{\circ}\text{C}$ (100.4°F). It is a regulated physiological response mediated by pyrogens acting on the hypothalamic thermostat. Fever is a symptom, not a disease, and its pattern, degree, and accompanying features guide the diagnosis.

CAUSE

Infectious (most common): Bacterial (UTI, pneumonia, sepsis, typhoid, TB), Viral (influenza, dengue, COVID-19, chikungunya — common in Tamil Nadu), Parasitic (malaria — must consider in India). Non-infectious: Drug fever, autoimmune (SLE, RA, vasculitis), malignancy (lymphoma — classic 'B symptoms': fever + night sweats + weight loss), post-surgical, deep vein thrombosis (low grade).

WHAT IT MEANS

Fever pattern carries diagnostic clues: Continuous (stays elevated) = bacterial pneumonia, typhoid. Remittent (fluctuates but never normal) = most bacterial infections. Intermittent (normal periods) = malaria (tertian/quartan patterns), abscesses. Saddle-back (two peaks) = dengue. Fever in India context: malaria, dengue, typhoid (enteric fever), and leptospirosis are high priority.

CAN LEAD TO

Untreated bacterial infection: sepsis, multi-organ failure, death. Febrile seizures in children (6 months–5 years) — frightening but usually benign. Hyperpyrexia ($>41^{\circ}\text{C}$): brain damage, disseminated intravascular coagulation (DIC). Untreated malaria (especially *P. falciparum*): cerebral malaria, death.

WHAT TO DO

Paracetamol (first choice) for symptomatic relief; ibuprofen also effective in adults. Do NOT use aspirin in children (Reye's syndrome risk). Blood tests: CBC (leucocytosis = bacterial, leucopenia = dengue/viral), Dengue NS1 antigen + IgM, Malaria blood smear/RDT, Blood culture for typhoid, Widal test, Urine culture if UTI suspected. Adequate hydration — fever increases insensible water losses. Antibiotics only for confirmed bacterial infections.

■ RED FLAGS — Seek Emergency Care Immediately:

- Fever $>41^{\circ}\text{C}$ (hyperpyrexia) — medical emergency.
- Fever + purpuric non-blanching rash — meningococcal septicaemia, call 112.
- Fever + severe headache + neck stiffness + photophobia — meningitis.
- Fever with severe thrombocytopenia (platelets $<50,000$) — dengue haemorrhagic fever.
- Fever + jaundice + dark urine in traveller — malaria or leptospirosis.
- Fever + hypotension + altered consciousness — septic shock — ICU emergency.

E2. Night Sweats

Night sweats are severe hot flushes occurring during sleep that drench clothing and bedding, not simply related to an overheated room. They are an important symptom requiring investigation when persistent.

CAUSE

Physiological: Menopause (most common cause in women 45–55), andropause (low testosterone in men). Infectious: TB (classic: fever + night sweats + weight loss), HIV, infective endocarditis, brucellosis, osteomyelitis, malaria. Malignancy: Lymphoma (Hodgkin's/Non-Hodgkin's — B symptoms), leukaemia. Medications: Antidepressants (SSRIs/SNRIs), tamoxifen, GnRH analogues, antipyretics. Endocrine: Hyperthyroidism, hypoglycaemia, pheochromocytoma (rare). Idiopathic.

WHAT IT MEANS

Night sweats in perimenopausal woman with hot flushes = vasomotor menopause. Night sweats + fever + weight loss in young person = TB or lymphoma until proven otherwise. Night sweats + lymph node enlargement = lymphoma (urgent evaluation). Night sweats on SSRI medication = drug side effect.

CAN LEAD TO

Untreated TB or HIV: disease progression and transmission. Untreated lymphoma: disease spread, reduced treatment efficacy. Menopausal symptoms affecting quality of life and cardiovascular health.

WHAT TO DO

CXR, TB sputum AFB, Mantoux/IGRA test if TB suspected. HIV test. FBC + film (lymphoma: lymphadenopathy + anaemia). Lymph node biopsy if enlarged nodes. PET-CT for staging lymphoma. TSH, fasting glucose, cortisol. HRT (hormone replacement therapy) for menopausal night sweats under physician guidance.

■ RED FLAGS — Seek Emergency Care Immediately:

- Night sweats + painless enlarging lymph nodes + weight loss — lymphoma evaluation urgently.
- Night sweats + chronic productive cough + haemoptysis + weight loss — TB.
- Drenching night sweats in person with HIV — opportunistic infection.

E3. Unexplained Weight Loss

Unintentional weight loss of >5% of body weight over 6–12 months, without deliberate dieting or exercise, is clinically significant and warrants thorough investigation. It is associated with serious underlying pathology in up to 25% of cases.

CAUSE

Malignancy (accounts for ~20–30%): Lung, GI, lymphoma, pancreatic, renal cell cancer. Endocrine: Hyperthyroidism, diabetes mellitus (uncontrolled), adrenal insufficiency. GI: Malabsorption (coeliac, Crohn's, chronic pancreatitis), IBD. Infections: HIV/AIDS, TB, chronic infections. Psychiatric: Depression, anxiety, anorexia nervosa. Cardiac: Heart failure (cardiac cachexia). Social: Food insecurity, dementia (forgetting to eat).

WHAT IT MEANS

Weight loss with increased appetite = hyperthyroidism or uncontrolled diabetes. Weight loss with decreased appetite + fatigue + night sweats = malignancy or TB. Weight loss with malodorous steatorrhea (fatty stools) = malabsorption. Weight loss with low mood and poor appetite = depression.

CAN LEAD TO

Delayed diagnosis of malignancy leads to advanced stage at presentation. Malnutrition, sarcopenia (muscle wasting), immunosuppression. Cardiac cachexia worsens heart failure prognosis.

WHAT TO DO

Comprehensive: FBC, CRP, LFT, RFT, TFTs, HbA1c, LDH, CXR. HIV test, TB screening. Anti-TTG IgA for coeliac. Tumour markers (CEA, CA19-9, PSA, AFP) as guided by clinical features. CT chest/abdomen/pelvis ('cancer screen CT') for unexplained weight loss in adults. Upper GI endoscopy + colonoscopy if GI symptoms. Psychiatric evaluation for eating disorders.

■ **RED FLAGS — Seek Emergency Care Immediately:**

- **Weight loss >10% with pain, haemoptysis, haematochezia — urgent oncology workup.**
- **Weight loss with night sweats and lymphadenopathy — lymphoma.**
- **Weight loss in elderly with cognitive decline — possible dementia with neglected nutrition.**

E4. Rash & Skin Changes

Skin rashes are one of the most visible manifestations of systemic and localised disease. The morphology, distribution, progression, and associated symptoms are key to diagnosis.

CAUSE

Infectious: Dengue (maculopapular then petechial — classic in Chennai), Chickenpox (vesicular), Shingles (dermatomal vesicular), Measles (maculopapular, head-to-toe), Scabies (intensely itchy). Allergic: Urticaria (hives), Drug rash (amoxicillin, sulfonamides, allopurinol), Contact dermatitis. Autoimmune: Lupus (butterfly/malar rash), Psoriasis (silvery plaques), Vitiligo, Dermatomyositis. Skin infections: Cellulitis, Tinea (ringworm), Impetigo. Systemic: Meningococcal purpura, Kawasaki disease (children).

WHAT IT MEANS

Non-blanching purpuric rash = blood in skin = meningococcal/vasculitis — emergency. Vesicular dermatomal rash with burning pain = shingles. Butterfly-shaped facial rash + joint pain + fatigue = SLE. Intensely itchy rash in web spaces + nocturnal itch = scabies.

CAN LEAD TO

Untreated cellulitis: necrotising fasciitis, sepsis. Untreated scabies: Norwegian (crusted) scabies in immunocompromised. Untreated SLE rash: systemic lupus progression — nephritis, CNS involvement. Stevens-Johnson Syndrome (drug reaction) — potentially fatal sloughing of skin.

WHAT TO DO

Dermatoscopy for skin lesions. Skin swab/scraping for infections. ANA, anti-dsDNA for suspected lupus. Biopsy for uncertain rashes. Discontinue causative medications for drug rashes. Antihistamines for urticaria; topical steroids for eczema; antivirals for shingles. Meningococcal vaccine for eligible contacts.

■ **RED FLAGS — Seek Emergency Care Immediately:**

- **Non-blanching purpuric rash + fever + neck stiffness — meningococcal — call 112.**
- **Widespread blistering and skin peeling after drug use — Stevens-Johnson Syndrome.**
- **Rash with mucosal involvement (mouth, eyes, genitals) — TEN/SJS emergency.**
- **Rapidly spreading red, hot, painful skin with fever — necrotising fasciitis.**

E5. Excessive Thirst & Urination (Polydipsia / Polyuria)

Polydipsia (excessive thirst) and polyuria (passing large volumes of urine) are classically linked and together form a diagnostic pair that strongly suggests metabolic or endocrine pathology.

CAUSE

Diabetes mellitus (Type 1 and 2 — hyperglycaemia causes osmotic diuresis). Diabetes insipidus (ADH deficiency or resistance — rare, passes >3L/day dilute urine). Primary hyperaldosteronism (Conn's syndrome — hypokalaemia). Hypercalcaemia (hyperparathyroidism, malignancy). Chronic kidney disease. Psychogenic polydipsia (psychiatric patients on antipsychotics). Medications: Lithium, diuretics.

WHAT IT MEANS

Thirst + polyuria + weight loss + fatigue = Type 1 diabetes until proven otherwise. Thirst + polyuria + recurrent infections + obesity = Type 2 diabetes. Polyuria with very dilute urine (not sweet) = diabetes insipidus. Polydipsia alone without polyuria = psychogenic.

CAN LEAD TO

Untreated Type 1 diabetes: DKA — medical emergency. Untreated Type 2 diabetes: HONK/HHNKS, progressive neuropathy, retinopathy, nephropathy, cardiovascular disease. Uncontrolled diabetes leads to blindness, amputations, kidney failure.

WHAT TO DO

Fasting blood glucose, HbA1c, random blood glucose. Urine glucose and ketones. Urine osmolality/specific gravity for DI. Serum calcium, PTH, potassium. Start metformin and lifestyle modification for Type 2. Insulin for Type 1 — must not be delayed. Referral to diabetologist for management and complication screening.

■ RED FLAGS — Seek Emergency Care Immediately:

- Polyuria + vomiting + abdominal pain + fruity breath in known/suspected T1DM — DKA — emergency.
- Severe dehydration with very high blood sugar + reduced consciousness — HONK syndrome.

PART F — MENTAL & HORMONAL SYMPTOMS

F1. Anxiety & Panic Attacks

Anxiety disorders are the most prevalent mental health conditions globally, affecting ~1 in 4 people at some point in their lives. Panic attacks are sudden, intense surges of fear accompanied by physical symptoms that can mimic a heart attack.

CAUSE

Psychological: GAD (generalised anxiety disorder), PTSD, social anxiety, OCD. Medical triggers: Hyperthyroidism, hypoglycaemia, pheochromocytoma, cardiac arrhythmias, mitral valve prolapse, vestibular disorders. Substance-related: Caffeine excess, alcohol withdrawal, stimulant drugs, benzodiazepine withdrawal. Genetic predisposition + environmental stressors.

WHAT IT MEANS

Panic attack: sudden onset heart racing, chest tightness, breathlessness, tingling, derealization, intense fear of dying — peaks in 10 minutes and resolves in 20–30 minutes. GAD: persistent, uncontrollable worry about multiple life domains for >6 months. Anxiety causing avoidance behaviour limits life significantly.

CAN LEAD TO

Agoraphobia (fear of situations where escape is difficult). Depression is highly comorbid with anxiety — 50% co-occurrence. Social isolation, occupational dysfunction. Substance misuse as self-medication. Physical health decline from chronic cortisol elevation.

WHAT TO DO

Rule out medical causes: ECG, TFTs, glucose, cortisol. Psychological: Cognitive Behavioural Therapy (CBT) — gold standard. Mindfulness-based therapies. Graded exposure. Pharmacological: SSRIs (sertraline, escitalopram) — first-line; SNRIs; Buspirone for GAD; short-term low-dose benzodiazepines for acute crisis only (risk of dependence). Lifestyle: Regular aerobic exercise reduces anxiety by 40–50%. Reduce caffeine. Diaphragmatic breathing, progressive muscle relaxation.

■ RED FLAGS — Seek Emergency Care Immediately:

- Panic attack indistinguishable from MI — always get ECG for first presentation.
- Anxiety with suicidal ideation — psychiatric emergency assessment.
- Severe agitation with confusion, hyperthermia, diaphoresis — possible thyroid storm or serotonin syndrome.

F2. Depression & Low Mood

Major Depressive Disorder (MDD) is characterised by persistent low mood and/or loss of pleasure (anhedonia) for >2 weeks, significantly impacting daily functioning. It is the leading cause of disability worldwide, affecting 280 million people.

CAUSE

Biopsychosocial model: Neurobiological (monoamine dysregulation — serotonin, norepinephrine, dopamine; HPA axis hyperactivity; neuroinflammation). Psychological: Negative cognitive schemas, chronic stress, trauma, loss. Social: Isolation, financial problems, relationship breakdown. Medical: Hypothyroidism, anaemia, chronic pain, cancer, post-stroke, post-MI. Medications: Beta-blockers, corticosteroids, isotretinoin, oral contraceptives (some). Postpartum depression (10–15% of new mothers). Seasonal Affective Disorder (SAD).

WHAT IT MEANS

Core symptoms: depressed mood, loss of interest/pleasure, fatigue, poor concentration, sleep disturbance (insomnia or hypersomnia), appetite changes, feelings of worthlessness, psychomotor changes, suicidal thoughts. PHQ-9 score >10 = moderate-severe depression requiring treatment.

CAN LEAD TO

Untreated depression: chronicity, recurrence, suicide risk (10–15% lifetime risk in severe cases). Cognitive impairment that may not fully reverse. Increased all-cause mortality — depression doubles cardiovascular mortality. Social and occupational devastation.

WHAT TO DO

PHQ-9 / BDI screening tools. Rule out medical causes (TFTs, B12, FBC). Psychotherapy: CBT, Interpersonal Therapy (IPT), Behavioural Activation. Antidepressants: SSRIs (first line — fluoxetine, sertraline, escitalopram); SNRIs (venlafaxine, duloxetine); TCAs (older, more side effects). Allow 4–6 weeks for full antidepressant effect. Continue 6–9 months after remission. Exercise prescription (150 min/week aerobic exercise — equivalent to antidepressant for mild-moderate). Safety planning if suicidal ideation present.

■ RED FLAGS — Seek Emergency Care Immediately:

- Active suicidal ideation with plan or intent — psychiatric emergency, do not leave alone.

- **Psychotic depression (hallucinations + severe depression) — inpatient treatment needed.**
- **Postpartum psychosis (rapidly developing in new mother) — psychiatric emergency.**

F3. Insomnia & Sleep Disturbance

Insomnia is difficulty initiating or maintaining sleep, or non-restorative sleep, at least 3 nights per week for >3 months, despite adequate opportunity, causing daytime impairment. It affects 10–30% of adults chronically.

CAUSE

Psychological: Anxiety (most common — hyperarousal), depression, stress, PTSD. Behavioural: Poor sleep hygiene, irregular schedule, daytime napping, screen use. Medical: Chronic pain, GERD, nocturia (BPH, diabetes), obstructive sleep apnoea, restless legs syndrome, hyperthyroidism, heart failure (orthopnoea). Substances: Caffeine, alcohol (disrupts REM), nicotine, stimulant medications. Medications: Beta-blockers, SSRIs, decongestants, corticosteroids. Circadian disruption: Shift work, jet lag.

WHAT IT MEANS

Difficulty falling asleep with racing thoughts = anxiety-driven insomnia. Early morning awakening (3–4 AM, unable to return to sleep) = classic depression pattern. Waking repeatedly to urinate = nocturia (prostate, diabetes, heart failure). Snoring + witnessed apneas + daytime sleepiness = OSA.

CAN LEAD TO

Chronic insomnia: 3x increased risk of depression; impaired immune function; cardiovascular disease; obesity and diabetes risk; cognitive impairment; accidents and injury (drowsy driving equivalent to drunk driving). Hypnotic dependence if sleeping pills used long-term.

WHAT TO DO

CBT-I (Cognitive Behavioural Therapy for Insomnia) — most effective long-term treatment. Sleep restriction therapy. Stimulus control. Relaxation techniques. Treat underlying conditions (pain, depression, OSA, GERD). Sleep hygiene: consistent sleep/wake time, dark/cool/quiet room, no screens 60 min before bed, no caffeine after 2 PM, no alcohol. Short-term medications: melatonin (0.5–5 mg for circadian disruption), low-dose doxepin, orexin antagonists (suvorexant). Avoid long-term benzodiazepines/z-drugs (tolerance and dependence).

■ RED FLAGS — Seek Emergency Care Immediately:

- **Daytime hypersomnia + sleep paralysis + sudden muscle weakness (cataplexy) — narcolepsy.**
- **Severe insomnia + confusion + rapid cognitive decline — possible fatal familial insomnia (rare).**
- **Sleep disorder causing severe daytime impairment affecting driving safety.**

F4. Hormonal Imbalance Signs

Hormones regulate virtually every bodily function. Imbalances — whether excess or deficiency — produce characteristic symptom clusters that often overlap with each other and with other conditions, making pattern recognition essential.

CAUSE

Thyroid dysfunction (hypo/hyperthyroidism — most common hormonal disorder). Insulin resistance / Type 2 diabetes. PCOS (polycystic ovary syndrome — affects 10% of women of reproductive age). Adrenal disorders: Cushing's syndrome (excess cortisol), Addison's disease (deficiency). Hypogonadism: Low testosterone (men), Oestrogen deficiency (perimenopause, premature ovarian insufficiency). Growth hormone excess (acromegaly) or deficiency. Prolactinoma (elevated prolactin — causes galactorrhoea, infertility).

WHAT IT MEANS			
CAN LEAD TO			
WHAT TO DO			
Hormone Issue	Key Symptoms	Key Tests	Primary Treatment
Hypothyroidism	Fatigue, weight gain, cold intolerance, constipation, dry skin, hair loss, bradycardia	TSH, Free T4	Levothyroxine
Hyperthyroidism	Weight loss, heat intolerance, palpitations, anxiety, tremor, diarrhoea, exophthalmos	TSH (low), Free T4, T3	Carbimazole, propylthiouracil, iodine, surgery
PCOS	Irregular periods, hirsutism, acne, weight gain, infertility, pelvic cysts	LH/FSH ratio, testosterone	Lifestyle, Metformin, OCP
Cushing's Syndrome	Central obesity, moon face, buffalo hump, purple stretch marks, hypertension, diabetes	24h UFC, dexamethasone suppression test	Surgery, medical (cause-dependent)
Addison's Disease	Fatigue, weight loss, hyperpigmentation, hypotension, salt craving	Cortisol, ACTH stimulation test	Hydrocortisone + fludrocortisone
Low Testosterone	Fatigue, low libido, ED, reduced muscle mass, osteoporosis	Total & free testosterone, LH, FSH	Testosterone replacement, specialist guidance
Menopause	Hot flushes, night sweats, vaginal dryness, mood changes, sleep issues	FSH, estradiol	HRT, lifestyle, CBT
Prolactinoma	Galactorrhoea, amenorrhoea, headache, visual field defects	Prolactin, MRI pituitary	Dopamine agonists (cabergoline)

PART G — QUICK REFERENCE TABLES

G1. Symptom-to-Condition Mapping

Symptom Cluster	Top Differential Diagnoses	Priority Level
Fever + body ache + thrombocytopenia (Indolent)	Dengue, Malaria, Typhoid	HIGH
Chronic fatigue + pale skin + breathlessness	Iron deficiency anaemia, B12 deficiency	MODERATE
Unilateral pulsating headache + nausea + photophobia	Migraine	MODERATE
Weight loss + night sweats + lymphadenopathy	Lymphoma, TB, HIV	HIGH
Polyuria + polydipsia + weight loss	Diabetes mellitus (T1 or T2)	HIGH
Crushing chest pain + sweating + jaw/arm radiation	Myocardial infarction	EMERGENCY
Sudden unilateral weakness + speech difficulty	Ischaemic stroke / TIA	EMERGENCY
RLQ pain + fever + anorexia + rebound tenderness	Acute appendicitis	EMERGENCY
Low mood + anhedonia + sleep change >2 weeks	Major depressive disorder	MODERATE–HIGH
Hot flushes + irregular periods + night sweats	Perimenopausal (45–55)	LOW–MODERATE
Fatigue + weight gain + cold intolerance + constipation	Hypothyroidism	MODERATE
Burning feet + reduced sensation + diabetes	Diabetic peripheral neuropathy	MODERATE
Sudden severe headache 'worst of life' + neck stiffness	Subarachnoid haemorrhage	EMERGENCY
Productive cough >2 weeks + night sweats + weight loss	Pulmonary tuberculosis	HIGH
Bloating + alternating bowel habit + cramping relieved by defecation	IBS	LOW
Palpitations + irregular pulse + stroke risk factors	Atrial fibrillation	HIGH
Joint pain + skin rash + fatigue + photosensitivity	Systemic lupus erythematosus	HIGH
Epigastric pain + worse lying flat + acid taste	GERD / Peptic ulcer disease	LOW–MODERATE
Shortness of breath + ankle oedema + orthopnoea	Congestive heart failure	HIGH
Severe dizziness + nausea + difficulty walking	Cerebellar stroke	EMERGENCY

G2. Urgent vs. Non-Urgent Symptom Guide

EMERGENCY (Call 112/999)	URGENT (Same-day GP)	ROUTINE (Appointment within 1–2 weeks)
Chest pain + sweating	Fever >3 days with worsening symptoms	Mild persistent fatigue
Sudden severe headache	Blood in urine (haematuria)	Chronic bloating
Stroke symptoms (FAST)	Unexplained weight loss	Mild headaches
Non-blanching rash + fever	Persistent vomiting/diarrhoea (dehydration)	Irregular periods
Breathing difficulty (SpO2<90%)	Severe abdominal pain	Mild joint aches
Sudden vision/speech loss	New neurological symptoms	Skin rash (stable, no systemic symptoms)
Unconsciousness/seizure	Palpitations with presyncope	Sleep problems
Suspected poisoning/overdose	Painful urination with fever/loin pain	Mood changes

EMERGENCY (Call 112/999)	URGENT (Same-day GP)	ROUTINE (Appointment within 1–2 wee
Severe allergic reaction	Acute painful swollen joint	Hair thinning
Suspected cauda equina	Coughing blood (haemoptysis)	Mild acid reflux

G3. Symptom Cluster Patterns for Common Diseases

Condition	Classic Symptom Cluster	Key Diagnostic Test
Type 2 Diabetes	Fatigue, polydipsia, polyuria, recurrent infections	HbA1c, fasting glucose
Hypothyroidism	Weight gain, fatigue, brain fog, hair loss, cold intolerance	TSH, Free T4
Iron Deficiency Anaemia	Fatigue, pallor, palpitations, breathlessness	BC, ferritin, transferrin
Fibromyalgia	Widespread muscle pain, fatigue, non-restorative sleep	Clinical diagnosis (e.g. IBS)
Dengue Fever	High fever, severe body pain, headache behind eyes	Dengue NS, haematology
Typhoid	Sustained fever, headache, abdominal pain, rose spots	Widal, blood culture
PCOS	Irregular/absent periods, acne, hirsutism, weight gain	Pelvic US, LH/FSH, testosterone
Rheumatoid Arthritis	Symmetric joint pain/swelling, morning stiffness	RF, anti-CCP, ESR/CRP
Irritable Bowel Syndrome	Abdominal cramping, bloating, altered bowel habits	Diagnosis of exclusion, ROME criteria
Major Depression	Low mood, anhedonia, fatigue, sleep disturbance	PHQ-9, clinical interview
Pulmonary TB	Productive cough >2 weeks, haemoptysis, night sweats	Sputum AFB, CXR, T-SPOT
Heart Failure	Dyspnoea on exertion/rest, orthopnoea, PND, leg swelling	BNP, echocardiography
Hyperthyroidism	Weight loss with increased appetite, heat intolerance	TSH (suppressed), Free T4
Migraine	Unilateral pulsating headache, nausea/vomiting	Clinical diagnosis; MRI if atypical
Vitamin B12 Deficiency	Fatigue, tingling hands/feet, balance problems	Serum B12, homocysteine, MCV

DISCLAIMER: This document is for educational and RAG/AI model training reference use only. It does not substitute professional medical advice, diagnosis, or treatment. Individual presentations vary widely. Always consult a qualified physician for personal health concerns. Emergency symptoms described require immediate professional intervention — do not delay calling emergency services. Sources: WHO, NHS Clinical Knowledge Summaries, UpToDate, NICE Guidelines, ICMR National Guidelines (India), Harrison's Principles of Internal Medicine, Oxford Handbook of Clinical Medicine.